$v = A\cos\omega t$

$$\frac{A^2}{2} = \frac{1}{T}\int_0 v^2(t)\,dt = V_{rms}^2$$

$$P = \frac{A^2}{2R}$$

Power and Phasor

$$P_{avg} = \frac{|V|}{\sqrt{2}} \cdot \frac{|I|}{\sqrt{2}} \cos(\phi) \qquad \phi = \theta_v - \theta_i$$

$$= V_{rms} \cdot I_{rms} \cdot \cos(\phi)$$

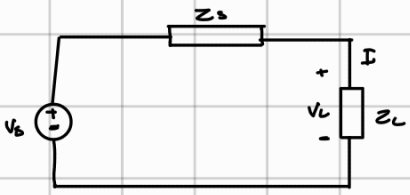
Complex power (VAR)

$S = P + jQ$ → reaktif güç (aktif güç: P)

$$= \frac{1}{2}(V \cdot I^*) = V_{rms} I^*_{rms}$$

$|S|$ = görünür güç

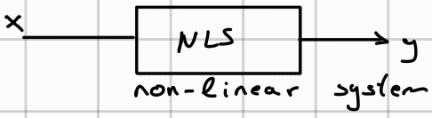
Maximum power



$$Z_s = Z_L^* \Rightarrow P_L = \frac{1}{8}\frac{|V_s|^2}{R_s}$$

① Harmonic distortion

$y = a_1 x + a_2 x^2 + a_3 x^2$



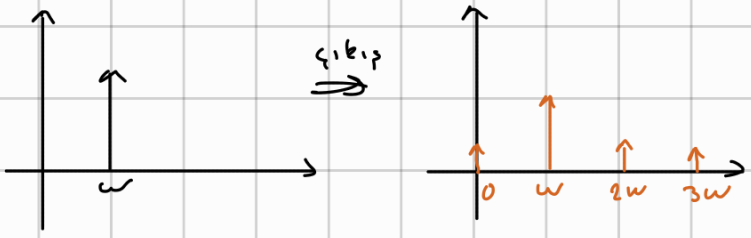
$x(t) = A\cos\omega t$

$$y(t) = a_1 A\cos\omega t + a_2 (A\cos\omega t)^2 + a_3 (A\cos\omega t)^3$$

$$(A\cos\omega t)^2 = \frac{A^2}{2}(1 + \cos 2\omega t)$$

$$(A\cos\omega t)^3 = \frac{A^3}{4}(3\cos\omega t + \cos 3\omega t)$$

$$y(t) = a_1 A\cos\omega t + a_2 \frac{A^2}{2}(1 + \cos(2\omega t)) + a_3 \frac{A^3}{4}(3\cos\omega t + \cos 3\omega t)$$



$$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4}\right)\cos\omega t + \frac{a_2 A^2}{2}\cos 2\omega t + \frac{a_3 A^3}{4}\cos 3\omega t$$

BPF $a_1 A + \frac{3a_3 A^3}{4}$ geçer

(2) Gain compression

$$V_{out} = \underbrace{-\sqrt{2k\frac{W}{L} I_D}\, R_D}_{a_1} V_{in} + \underbrace{\frac{2k^{3/2}}{8\sqrt{I_D}}\, R_D}_{\frac{3a_3}{4}} V_{in}^3$$

a_1, a_3 have different signs

Most of the time $a_1 \cdot a_3 < 0$

$$Gain = \frac{V_{out}(\text{fundemental})}{V_{in}} = \frac{a_1 A + \frac{3a_3 A^3}{4}}{A}$$

$$V_{out} = a_1 V_{in} + \frac{3a_3}{4} V_{in}^3$$

$$Gain = \frac{V_{out}}{V_{in}} = \frac{a_1 A + \frac{3a_3 A^3}{4}}{A}$$

$$\Rightarrow gain = a_1 + \frac{3}{4} a_3 A^2$$

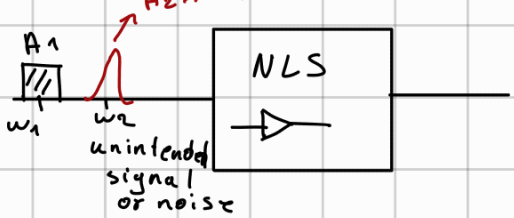
1) $a_1 > 0$, $a_3 < 0 \rightarrow Gain = a_1 - \frac{3}{4}|a_3| A^2 \rightarrow V_{in}$ gain girişe bağlı

2) $Gain \propto A^2$

girişin genliği çok yükselirse kazanç düşer

(3) Desensitization

$A_2 \gg A_1$ olsun

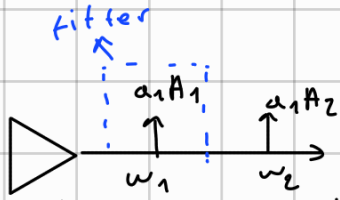


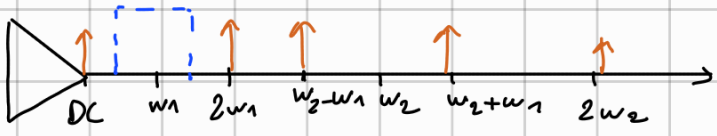
$$x(t) = \underbrace{A_1 \cos\omega_1 t}_{\text{desired input signal}} + \underbrace{A_2 \cos\omega_2 t}_{\text{unwanted signal}}$$

$$y = a_1 x + a_2 x^2 + a_3^3$$

$$y(t) = a_1 A_1 \cos\omega_1 t + a_1 A_2 \cos\omega_2 t + a_2 \left[A_1^2 \cos^2\omega_1 t + A_2^2 \cos^2\omega_2 t \right] + 2 a_2 A_1 A_2 (\cos\omega_1 t)(\cos\omega_2 t)$$

$\cos(\omega_1+\omega_2)t + \cos(\omega_2-\omega_1)t$





$$+a_3\left[A_1^3\cos^3\omega_1 t + A_2^3\cos^2\omega_2 t + 3A_1\cos\omega_1 t \cdot \underbrace{A_2^2\cos^2\omega_2 t}_{1+\cos 2\omega_2 t} + 3A_2\cos\omega_2 t \cdot A_1^2\cos^2\omega_1 t\right]$$

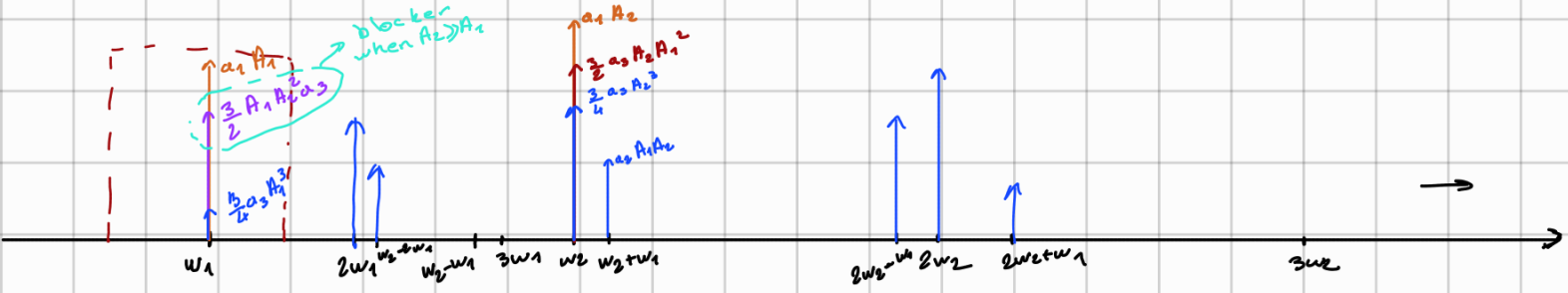
$$\frac{3}{4}a_3A_1^3\cos\omega_1 t + \frac{A_1^3}{4}\cos 3\omega_1 t + \cos\omega_2 t + \cos 3\omega_2 t + \cos\omega_1 t + \cos(2\omega_1-\omega_2)t + \cos(2\omega_2+\omega_1)t + \cos\omega_2 t + \cos(2\omega_1-\omega_2)t + \cos(2\omega_1+\omega_2)t$$

$n\omega_1 + m\omega_2$

$n + |m| = 3$

gelen singal

$$\frac{3}{4}a_3A_1^3\cos\omega_1 t + \frac{3A_1A_2^2}{2}a_3\cos\omega_1 t$$

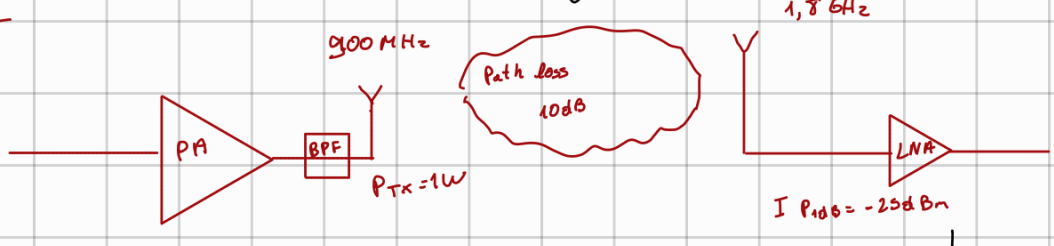


ω_1 frekansında olan singal: $a_1A_1\cos\omega t + \frac{3}{4}a_3A_1^3\cos\omega_1 t + \frac{3A_1A_2^2}{2}a_3\cos\omega_1 t$

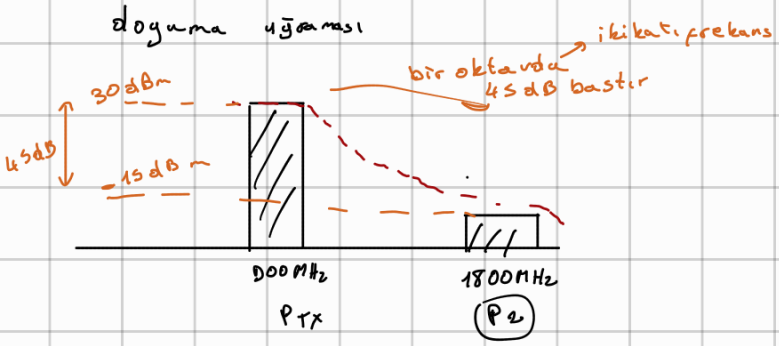
Gain: $\dfrac{(a_1A_1 - \frac{3}{2}a_3A_1A_2^2)\cos\omega_1 t}{A_1\cos\omega_1 t}$ $\Big\}$ $A_2 \uparrow\uparrow$

No signal appears at the output

Ex



Vericinin ikinci harmoniğine kadar bastırılmalıdır ki alıcı doğuma uğramasın



$P_2 - 10dB < P_1dB$

$P_2 - 10 < -25dB$

$P_2 < -15$ dBm

0 dBm 1 mwatt

10 dBm 10 mwat

$10\log\left(\frac{1W}{10^{-3}}\right) = 30$ dBm

en az 45dB bastırılmalıdır.

④ Intermodulation

$A_{IM3} = \frac{3}{4} a_3 A_1^2 A_2 \cos(2\omega_1 - \omega_2)t$

$A_1 = A_2 = A \rightarrow A_{IM3} = \frac{3}{4} a_3 A^3$

if system linear $\rightarrow a_1 A$

Intermodulation products $\rightarrow \frac{3}{4} a_3 A^3$

$IM3 = \frac{\frac{3}{4} a_3 A^3}{a_1 A} = \frac{3}{4} \left| \frac{a_3}{a_1} \right| A^2$

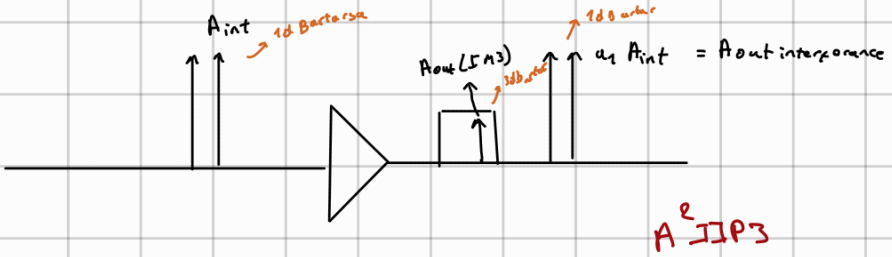
$A_{IIP3} \Rightarrow 20\log(|a_1| A_{IIP3}) = 20\log\left(\frac{3}{4}|a_3| A_{IIP3}^3\right)$

$$\boxed{A_{IIP3} = \sqrt{\frac{4}{3}\left|\frac{a_1}{a_3}\right|}}$$

$A_{1dB} = \sqrt{0{,}145 \left|\frac{a_1}{a_3}\right|}$

$A_{IIP3}(dB) - A_1(dB) = 9dB$

if $A_{IIP3} \uparrow$ more linear system



$$\frac{A_{out,int}(\omega_1,\omega_2)}{A_{out}(IM3)} = \frac{a_1 A_{int}}{\frac{3}{4} a_3 A^3_{int}} = \frac{a_1}{\frac{3}{4} a_3 A^2_{int}} = \frac{A^2_{IIP3}}{A^2_{int}}$$

A^2_{IIP3}

$$A^2_{IIP3} = \frac{A_{out,int}(\omega_1,\omega_2)}{A_{out}(IM3)} A^2_{int}$$

$$20\log(A_{IIP3}) = 20\log(A_{int}) + \frac{1}{2}\left[20\log A_{out,in}(\omega_1,\omega_2) - 20\log A_{out}(IM3)\right]$$

Ex

$P = \frac{U^2}{R}$

$$10\log\left(\frac{P_{IIP3}}{10^{-3}}\right) = -10\,dBm$$

W

$\Rightarrow P_{IIP3} = 10^{-4} W = 0{,}1 mW$

$P_{IIP3} = \frac{A^2_{IIP3}}{2}$ $\quad \sqrt{A^2_{IIP3}} = \sqrt{10^{-4} \cdot 50}$

$A^2_{IIP3} = 70 mV$

$$A^2_{IIP3} = \frac{A_{out,int}(w_1,w_2)}{A_{0,IM3}} \cdot A^2_{in,int}$$

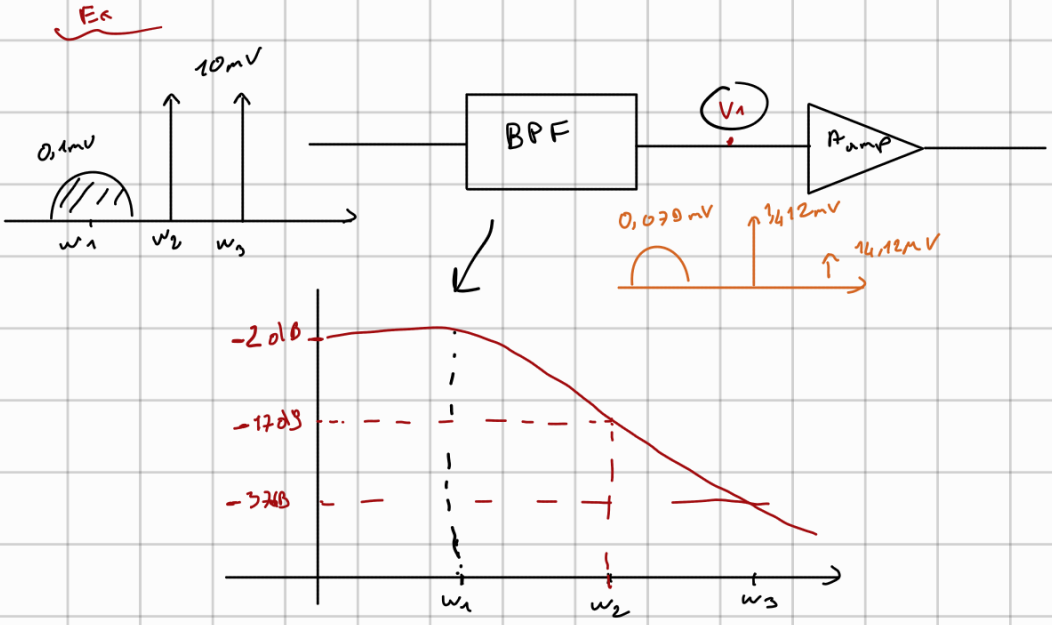
$$A^2_{IIP3} = \frac{a_1 A_{in,int}}{A_{0,IM3}} \cdot A^2_{in,int} = \frac{a_1 A^3_{in,int}}{A_{0,im3}}$$

$A_{0,S}$

$$A^2_{IIP3} = \frac{a_1 A^3_{in,int}}{A_{0,im3}} \cdot \frac{A_{in,S}}{A_{in,S}} = \boxed{\frac{A_{0,S}}{A_{0,IM}}} \cdot \frac{A^3_{in,int}}{A_{in,S}}$$

$$\frac{A_{0,S}}{A_{0,im3}} = A^2_{IIP3} \cdot \frac{A_{in,S}}{A^3_{in,int}} = (70mV)^2 \frac{1mV}{(1mV)^3} = 4,9 \longrightarrow \simeq 13,8\ dO$$

Ex



Compute IIP3 of the amplifier such that IM3 product falling at w1 is 20 dB below the desired signal ?

↳ $A_{0,IM3}$??

↳ $A_{0,S}$

$$\frac{A_{0,S\,dB}}{A_{0,IM3\,dB}} = 20dB \Rightarrow \frac{A_{0,S}}{A_{0\,IM3}} = 10$$

$$20\log A_{sig}(w_1) - 20dB = 20\log A_{sig_{BPF}}(w_1)$$

$$20\log \frac{A_{sig}}{A_{sig_{BPF}}} = 2 \Rightarrow \frac{A_{sig}}{A_{sig_{BPF}}} = 10^{1/10} = \frac{0,1 \cdot 10^{-3} V}{A_{sig_{BPF}}}$$

$$A_{sig_{BPF}}(w_1) = 0,079\,mV = V_1$$

$$20\log \underbrace{A_{int}(w_2)}_{10mV} - 17dBB = 20\log A_{int_{BPF}}(w_2) \Rightarrow A_{int_{BPF}}(w_2) = 1,42mV \ (V_2)$$

$$\dots A_{int\,BPF}(w_3) = 0,141\,mV \ (V_3)$$

$$A_{o\,sig}\ (dB) - A_{o\,IM3}\ (dB) = 20\,dB$$

$$A_{o,s} = a_1 V_1$$

$$A_{o,IM3} = \frac{4}{3} a_3 A_2^2 A_3 = \frac{3}{4} a_3 V_2^2 V_3$$

$$\rightarrow 20\log|a_1 V_1| - 20\log\left|\frac{3}{4} a_3 V_2^2 V_3\right| = 20\log 10$$

$$\frac{a_1 V_1}{\frac{3}{4} a_3 V_2^2 V_3} = 10 \Rightarrow \frac{a_1}{a_3} = 26{,}6 \cdot 10^{-6}$$

$$A_{IIP3} = \sqrt{\frac{4}{3}\left|\frac{a_1}{a_3}\right|} = \sqrt{\frac{4}{3}\left|26{,}6 \cdot 10^{-6}\right|} = 5{,}95\,mV \approx 15{,}5\,dBm$$

Ex : $a_1 = 10$, $A_{in,int} = 10\,mV$, $A_{IIP3} = 80\,mV$ } $IM3 = ?$, $A_{o_IM3} = ?$

$$A_{IIP3} = \sqrt{\frac{4}{3}\left|\frac{a_1}{a_3}\right|}$$

$$80 \cdot 10^{-3} = \sqrt{\frac{4}{3} \cdot \frac{10}{a_3}}$$

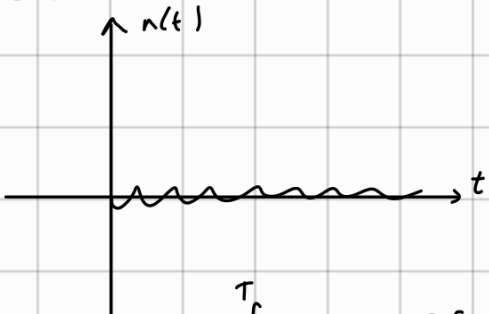
$$a_3 = \frac{4 \cdot 10}{3 \cdot (80 \cdot 10^{-3})^2} = \boxed{2083}$$

$$IM3 = \frac{\frac{3}{4} a_3 A^3}{a_1 A} = \frac{\frac{3}{4}(2083)(10 \cdot 10^{-3})^2}{10} = \boxed{0{,}0156}$$

$$A_{o,IM3} = \frac{3}{4} a_3 A^3 = 0{,}001562\ V$$

$$= 1{,}562\,mV$$

Noise

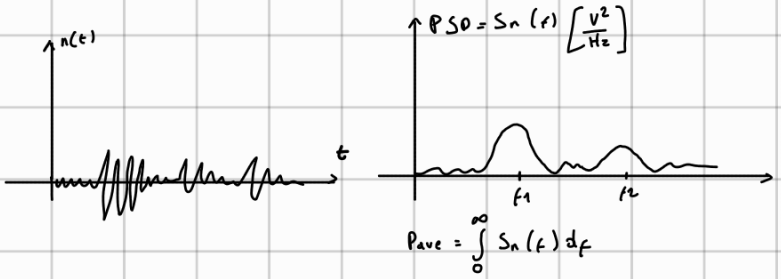


$$P_{ave} = \lim_{T\to\infty} \frac{1}{T}\int_0^T n^2(t)\,dt = E[n^2(t)]$$

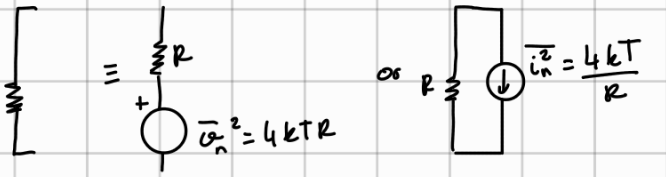
$$\sigma^2 = E[n^2] - E^2[n]$$

P_{AC} : $E[n^2]$ = P_{ave} ; $E^2[n]$ = P_{DC}

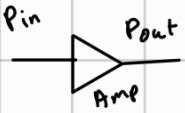
$$P_{ave} = P_{AC} + P_{DC}$$



$$P_{ave} = \int_0^\infty S_n(f)\,df$$

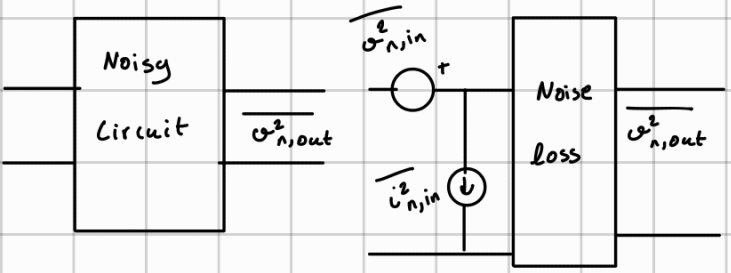


Input referred Noise



$$\Rightarrow P_{out} = P_{in}\cdot G$$

$$N_{out} = N_{in}\cdot G + N_A$$

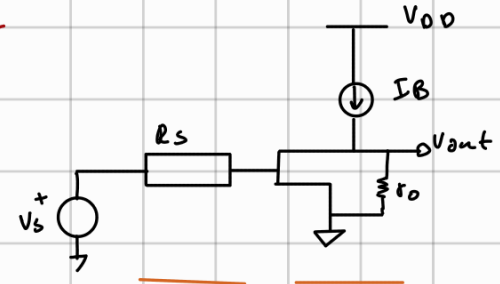


$$A_v = \frac{v_{out}}{v_{in}}$$

$$A_{vt} = \frac{v_{out}}{v_s} = \underbrace{\frac{v_{out}}{v_{in}}}_{A_v}\cdot\frac{v_{in}}{v_s}$$

How to find input referred Noise

Ex:



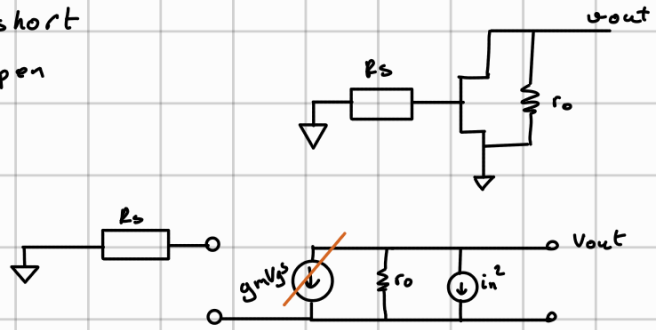
$$\overline{v_{n,out_1}^2} + \overline{v_{n,out_2}^2} + \overline{v_{n,out_3}^2} + \dots = \overline{v_{n,out}^2}_{total}$$

$$\overline{v_{n,input}^2} = \frac{\overline{v_{n,out(total)}^2}}{A_{vt}^2}$$

① Cancel sources (gerilim k. kısa, akım k. açık devre)

$v_s \rightarrow$ short

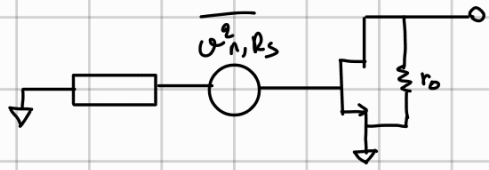
$I_B \rightarrow$ open



$$\Rightarrow \overline{v_{out,1}^2} = \overline{i_n^2}\cdot r_o^2 \qquad \overline{i_n^2} = 4kT\gamma g_m$$

$$\overline{v_{out,1}^2} = r_o^2\,4kT\gamma g_m$$

②



$$\overline{v_{n,out_2}^2} = \overline{v_{n,Rs}^2}\underbrace{(g_m r_o)^2}_{\text{Transistor gain}} = 4kTR_s(g_m r_o)^2$$

③ Sum of squares

$$\overline{v_{n,out(total)}^2} = \overline{v_{n,out_1}^2} + \overline{v_{n,out_2}^2}$$

$$= r_o^2\,4kT\gamma g_m + (g_m r_o)^2 kTR_s$$

④ $$\overline{v_{in,n}^2} = \frac{\overline{v_{n,out(total)}^2}}{A_{vt}^2} = \frac{r_o^2\,4kT\gamma g_m + (g_m r_o)^2\,4kTR_s}{(g_m r_o)^2}$$

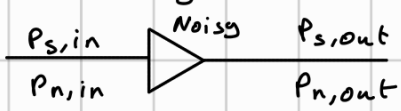
$$\overline{v_{in,n}^2} = \underbrace{\frac{4kT\gamma}{g_m}}_{\substack{\text{amplifierden}\\ \text{girişe gansigan}\\ \text{gürültü}}} + \underbrace{4kTR_s}_{\substack{\text{kaynağın}\\ \text{gürültüsü}}}$$

To decrease $\overline{v_{in,n}^2}$, increase g_m

genellikle amplifierlar 10, 15 dB üstü tasarlanır ki gürültüsü az olsun

Noise Figure (NF)
Noise Factor (F)

$NF = 10 \log(F)$



$$SNR_{in} = \frac{P_{s,in}}{P_{n,in}}$$

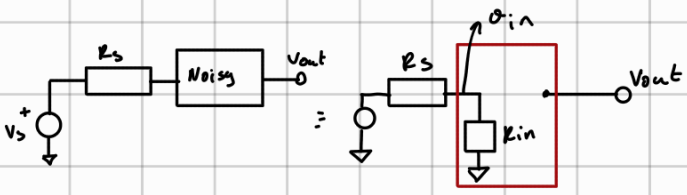
$$SNR_{out} = \frac{P_{s,out}}{P_{n,out}} = \frac{P_{s,out}}{P_{n,in} \cdot G + P_{n,A}}$$

$SNR_{in} > SNR_{out}$

$$F = \frac{SNR_{in}}{SNR_{out}} > 1$$

$$NF = 10 \log F = SNR_{in,dB} - SNR_{out,dB} > 0\,dB$$

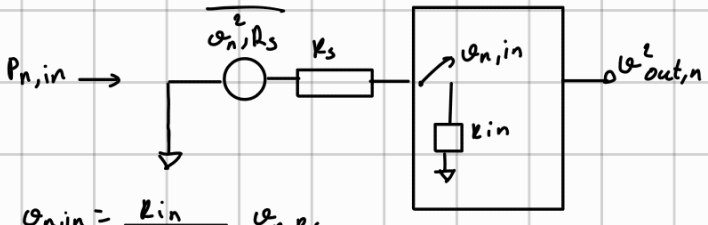
good LNA → NF : 0 ~ 2dB



$$SNR_{in} = \frac{P_{s,in}}{P_{n,in}}$$

$$P_{s,in} \rightarrow v_{in} = \frac{R_{in}}{R_{in}+R_s} v_s$$

$$P_{s,in} = \frac{v_{in}^2}{R_{in}} = \frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \cdot v_s^2}{R_{in}}$$



$$v_{n,in} = \frac{R_{in}}{R_{in}+R_s} v_{n,R_s}$$

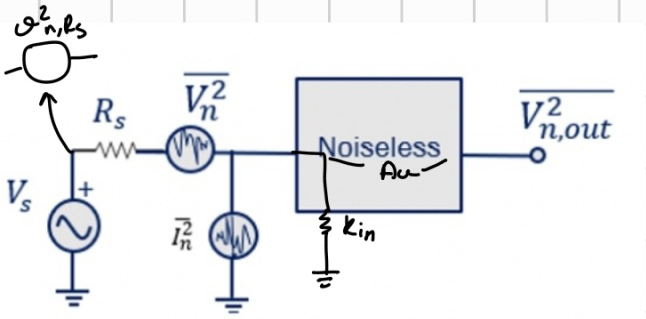
$$P_{n,in} = \frac{\overline{v_{n,in}^2}}{R_{in}} = \frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \overline{v_{n,R_s}^2}}{R_{in}}$$

$$SNR_{in} = \frac{\frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \cdot v_s^2}{R_{in}}}{\frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \overline{v_{n,R_s}^2}}{R_{in}}} = \frac{v_s^2}{\overline{v_{n,R_s}^2}}$$ ← sadece kaynağa bağlı

$$SNR_{out} = \frac{P_{s,out}}{P_{n,out}}$$

What is $SNR_{out} = ?$ $= \frac{P_{s,out}}{P_{n,out}}$

$$P_{s,out} = \left[V_s \left(\frac{R_{in}}{R_{in}+R_s} \right) \right]^2 \cdot A_v^2 = v_s^2 \left(\frac{R_{in}}{R_{in}+R_s} \right)^2 A_v^2$$



① Noise coming from R_s ($\overline{v_n^2} = 0$, $\overline{I_n^2} = 0$)

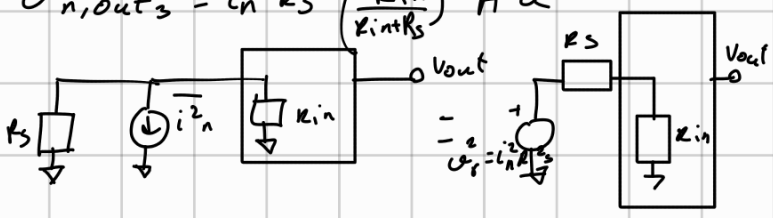
$$\overline{v_{n,out_1}^2} = \overline{v_{n,R_s}^2} \left(\frac{R_{in}}{R_s+R_{in}} \right)^2 A_v^2$$

② Noise coming from $\overline{v_n^2}$ ($\overline{v_{n,R_s}^2} = 0$, $\overline{I_n^2} = 0$)

$$\overline{v_{n,out_2}^2} = \overline{v_n^2} \left(\frac{R_{in}}{R_s+R_{in}} \right)^2 A_v^2$$

③ Noise coming from $\overline{i_n^2}$ ($\overline{v_n^2} = 0$, $\overline{v_{n,R_s}^2} = 0$)

$$\overline{v_{n,out_3}^2} = \overline{i_n^2} R_s^2 \left(\frac{R_{in}}{R_{in}+R_s} \right)^2 A_v^2$$



$$SNR_{out} = \frac{V_s^2}{\overline{v_{n,R_s}^2} + \overline{v_n^2} + \overline{i_n^2} R_s^2}$$

$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{V_s^2 / \overline{v_{n,R_s}^2}}{V_s^2 / \overline{v_{n,R_s}^2} + \overline{v_n^2} + \overline{i_n^2} \cdot R_s^2}$$

$$F = \frac{\overline{v_{n,R_s}^2} + \overline{v_n^2} + \overline{i_n^2} R_s^2}{\overline{v_{n,R_s}^2}}$$

$$F = 1 + \frac{\overline{v_n^2} + \overline{i_n^2} R_s^2}{\overline{v_{n,R_s}^2}}$$

- $\overline{v_n^2} + \overline{i_n^2} R_s^2$ → input referred noise
- $\overline{v_{n,R_s}^2}$ → source noise

$$A_v = \frac{v_{out}}{v_{in}}$$

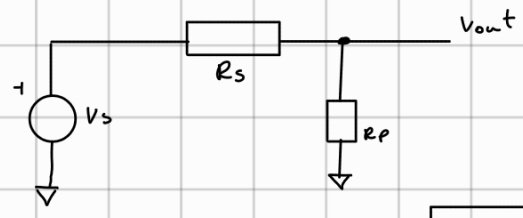
$$A_{vt} = \frac{v_{out}}{v_s} = \frac{v_{out}}{v_{in}} \cdot \frac{v_{in}}{v_s}$$

$$F = \frac{A_{vt}^2}{A_{vt}^2} \left(\frac{\overline{v_{n,R_s}^2} + \overline{v_n^2} + \overline{i_n^2} \cdot R_s^2}{\overline{v_{n,R_s}^2}} \right)$$

$$= \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{\overline{v_{n,R_s}^2}}$$

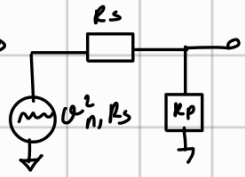
$$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{4kTR_s}$$

Ex:



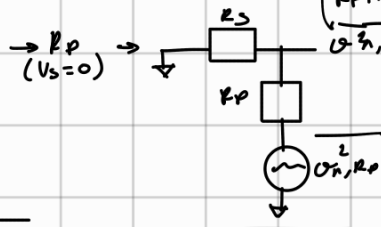
$$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{4kTR_s} \quad , \quad \boxed{A_{vt} = \frac{R_p}{R_s+R_p}}$$

$\overline{v_{n,out}^2}$ (total) $\rightarrow R_s$ ($V_s=0$) $\rightarrow$



$$\overline{v_{n,out_1}^2} = \left(\frac{R_p}{R_p+R_s}\right)^2 \cdot \overline{v_{n,R_s}^2}$$

$\rightarrow R_p$ ($V_s=0$) $\rightarrow$



$$\overline{v_{n,out_2}^2} = \left(\frac{R_s}{R_s+R_p}\right)^2 \overline{v_{n,R_p}^2}$$

$$\overline{v_{n,out\ (total)}^2} = \left(\frac{R_s}{R_p+R_s}\right)^2 \underbrace{\overline{v_{n,R_p}^2}}_{4kTR_p} + \left(\frac{R_p}{R_p+R_s}\right)^2 \underbrace{\overline{v_{n,R_s}^2}}_{4kTR_s}$$

$$= 4kT\,\frac{R_sR_p^2+R_pR_s^2}{(R_p+R_s)^2} = 4kT\,\frac{R_sR_p(R_s+R_p)}{(R_s+R_p)^2}$$

$$\overline{v_{n,out\ (total)}^2} = 4kT\,\underbrace{\frac{R_sR_p}{R_s+R_p}}_{R_s//R_p}$$

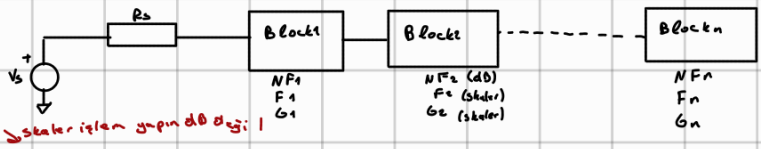
$$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2}\,\frac{1}{4kTR_s} = \frac{R_s+R_p}{R_p} = 1+\frac{R_s}{R_p}$$

$R_p \Uparrow \rightarrow F \approx 1$ gürültüsüz

$R_p \gg R_s \rightarrow$ No noise effect comming from R_p

Çalışma sorusuna bak
Ders 6

Noise in Cascaded Stages



Sıralar işlem yapmadığı için

Friis' Formülü

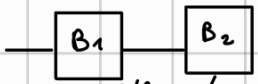
$$F(total) = 1 + (F_1-1) + \frac{F_2-1}{G_1} + \frac{F_3-1}{G_1G_2} + \frac{F_4-1}{G_1G_2G_3} + \cdots + \frac{F_n-1}{G_1G_2\cdots G_{n-1}}$$

$$F(total) = F_1 + \frac{F_2-1}{G_1} + \frac{F_3-1}{G_1G_2}$$

$\approx$ >10dB

Dominant F mutlaka girişte düşük gürültülü ve NF düşük olan amplifier seçilmeli

Ex



$G_1 = 10dB$ $NF_1 = 2dB$ $G_2 = 4dB$ $NF_2 = 10dB$

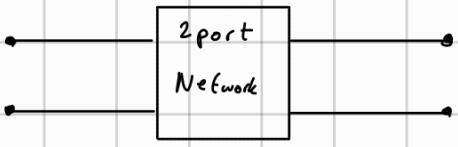
$NF_t = ??$ $F_1 = 10^{\frac{2}{10}} = 1,58$ $F_2 = 10^{\frac{10}{10}} = 10$ $G_1 = 10$

$$F_t = F_1 + \frac{F_2-1}{G_1} = 1,58 + \frac{10-1}{10} = 2,48$$

$$NF_t = 10\log F_t = 3,95\,dB$$

Noise in passive Reciprocal Circuit

Reciprocity



Reciprocity ← pasif devrelerde olabiliyor

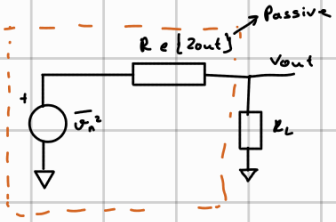
$$S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \qquad S_{12} = S_{21}$$

S_{11} → Return loss

S_{22} → Output Return loss

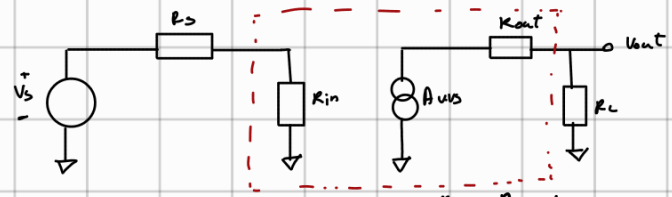
S_{21} → Gain

S_{12} → Isolation



$$\overline{v_n^2} = 4kT\,\underbrace{Re\{Z_{out}\}}_{R_{out}}$$

$$\overline{v_{n^2,out}} = \overline{v_n^2}\left(\frac{R_L}{R_L+R_{out}}\right)^2 = 4kTR_{out}\left(\frac{R_L}{R_L+R_{out}}\right)^2$$



$R_s = R_{in} \Rightarrow \frac{R_{in}}{R_s + R_{in}} = \frac{1}{2}$

Input matched

$R_L = R_{out}$

Output matcheol

$P_{in} = \frac{v_{in}^2}{R_{in}} = \frac{v_s^2}{4 R_{in}}$

$v_{in} = V_s \left(\frac{R_{in}}{R_s + R_{in}} \right)$

$P_{out} = \frac{(A_v \cdot V_s)^2}{2(R_L + R_{out})} = \frac{(A_v V_s)^2}{4 R_L}$

$\frac{P_{out}}{P_{in}} = G$

$G = \frac{\frac{(A_v V_s)^2}{4 R_L}}{\frac{V_s^2}{4 R_{in}}} = A_v^2 \frac{R_s}{R_L} = G$

zaygınlama

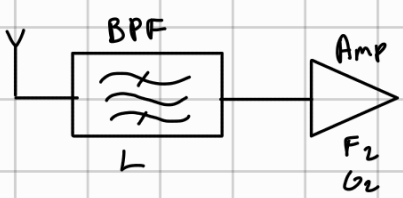
$L = \frac{1}{G} = \frac{1}{A_v^2} \cdot \frac{R_L}{R_s}$

$F = \frac{v_{n,out}^2}{A_{vt}^2} \cdot \frac{1}{4kTR_s} = \frac{4kTR_{out} \left(\frac{R_L}{R_L + R_{out}} \right)^2}{\left(\frac{A_v}{2} \right)^2} \cdot \frac{1}{4kTR_s}$

$= \frac{4kTR_L}{4} \cdot \frac{1}{\left(\frac{A_v}{2} \right)^2} \cdot \frac{1}{4kTR_s} \Rightarrow \frac{1}{A_v^2} \cdot \frac{R_L}{R_s} = F$

$A_{vt} = \left(\frac{V_o}{V_{in}} \right)_{A_v} \left(\frac{V_{in}}{V_s} \right)_{1/2} = \frac{A_v}{2}$

Ex:



$F_t = ?$

$F_t = F_1 + \frac{F_2 - 1}{G_1} = L + \frac{F_2 - 1}{\frac{1}{L}} = L + LF_2 - L = LF_2 = F_{total}$

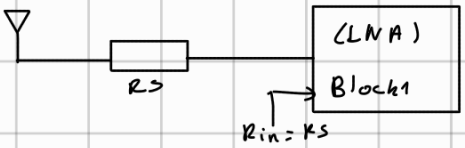
$F_1 = L$

$G_1 = \frac{1}{L}$

$NF_t = L_{dB} + NF_2$

Noise figürü artırır

Noise power in Rx input



$\overline{v_{in,n}^2} = \left(\frac{R_{in}}{R_{in} + R_s} \right)^2 \cdot \overline{v_s^2} = kTR_s$

$\left(\frac{R_{in}}{R_{in} + R_s} \right)^2 = \frac{1}{4}$

$\overline{v_s^2} \downarrow 4kTR_s$

$\frac{\overline{v_{n,in}^2}}{R_{in}} = \frac{kTR_s}{R_{in}} = kT$ (1Hz BW)

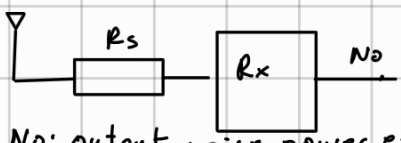
$R_{in} = R_s$

1Hz'lik bantta kT kadar gürültü gücü buluruz

$P_{n,in} = kTB$ (Input noise power in bandwith B)

Receive Noise Floor

Noise floor = Receiver input equivalent noise



No: output noise power excluding Rs kaynak dürünülmediğinde görülen

$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{4kTR_s}$

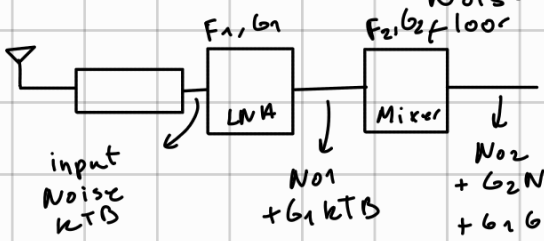
$F = \frac{\text{Total output noise power including } R_s}{G(\text{Noise power from } R_s \text{ only})}$

$= \frac{N_o + GkTB}{GkTB} = F$

???

Input noise $= kTB + \frac{N_o}{G} = FkTB$

Noise floor

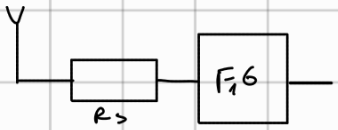


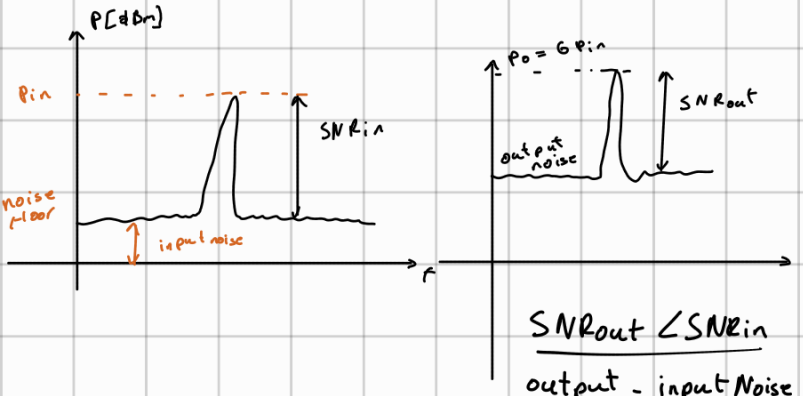
Receiver noise Floor

$$F = \frac{No + G\,kTB}{G} \cdot \frac{1}{kTB}$$

$$\text{Noise floor} = F\,kTB = \frac{No}{G} + kTB$$

Receiver SNR & Noise



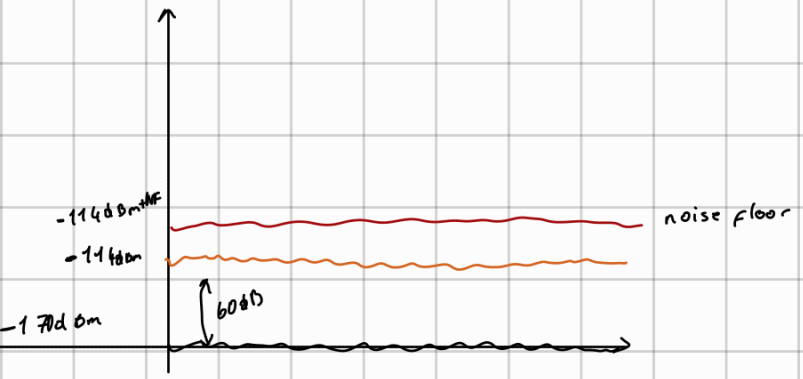


$SNR_{out} < SNR_{in}$

output noise = input Noise + NF

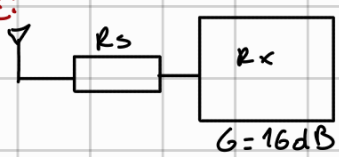
$$\text{Noise floor} = FkTB = \underbrace{10\log F}_{NF} + \underbrace{10\log kT}_{-174\,\frac{dBm}{Hz}} + 10\log B$$

$$\text{Noise floor}_{(dB)} = -174\,\frac{dBm}{Hz} + 10\log B + NF$$



Output referred = $FkTB\,G = NF - 174 + 10\log B + 10\log G$ total noise

Ex:



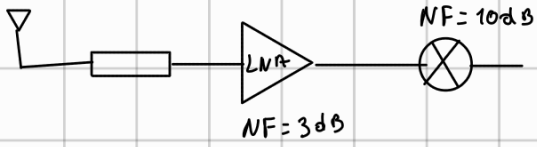
G = 16 dB
NF = 10 dB
B = 10 MHz

$$\text{noise floor} = FkTB = -174\,\frac{dBm}{Hz} + 10\log(10\cdot10^6) + 10$$

$$= -174 + 70 + 10 = -94\,dBm$$

$$N_{out,t} = -94 + 16 = -78\,dBm$$

Ex:



Calculate NF total

$$F = F_1 + \frac{F_2 - 1}{G_1}$$ (Skaler almali)

$$= 10^{3/10} + \frac{10^{12/10} - 1}{10^{12/10}} = 2{,}57$$

$$NF = 10\log(F) = 10\log(2{,}57) = 4{,}1\,dB$$

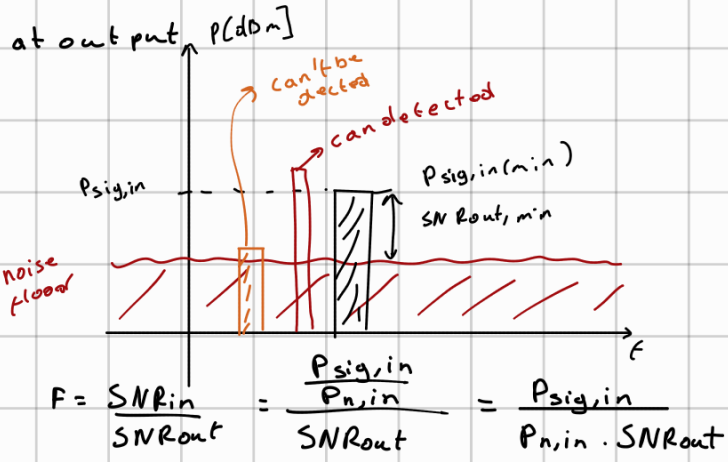
Calculate Noise floor B = 5MHz

$$\text{Noise floor} = -174\,\frac{dBm}{Hz} + 10\log(5\cdot10^6) + 4{,}1$$

$$= -102{,}9\,dBm$$

Sensitivity and Dynamic Range

Sensitivity is the minimum with can be detected at output



$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{\frac{P_{sig,in}}{P_{n,in}}}{SNR_{out}} = \frac{P_{sig,in}}{P_{n,in}\cdot SNR_{out}}$$

$$P_{sig,in} = \underbrace{P_{n,in}}_{kTB \text{ if } R_s = R_{in}} \cdot F \cdot SNR_{out}$$

$$P_{sig,in} = \underbrace{kTB \cdot F}_{\text{Noise floor}} \cdot SNR_{out}$$

$$P_{sig,in}(min)\Big|_{dB} = \text{noise floor}_{dB} + SNR_{out\,dB}$$

$$\underbrace{P_{sig,in}(min)\Big|_{dB}}_{\text{sensitivity}} = -174\,\frac{dBm}{Hz} + 10\log B + NF + SNR_{out}\Big|_{dB}$$

Ex: A GSM receiver has $SNR_{out(min)} = 12dB$

channel BW = 20 kHz

NF = 9 dB

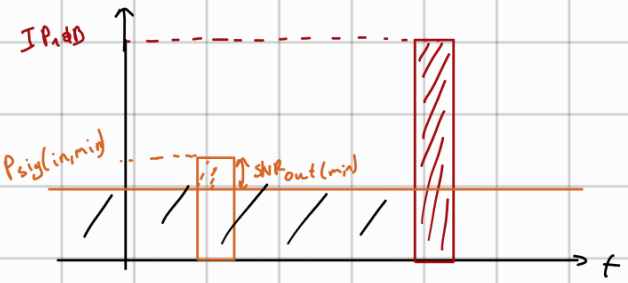
what is tthe minimum acceptable signal power?

$$P_{sig,in}(min) = -174 \frac{dBm}{Hz} + 10\log\underbrace{(20\cdot10^3)}_{43} + 9 + 12$$

$$= -112 + 12 = -100\ dBm$$



Dynamic Range

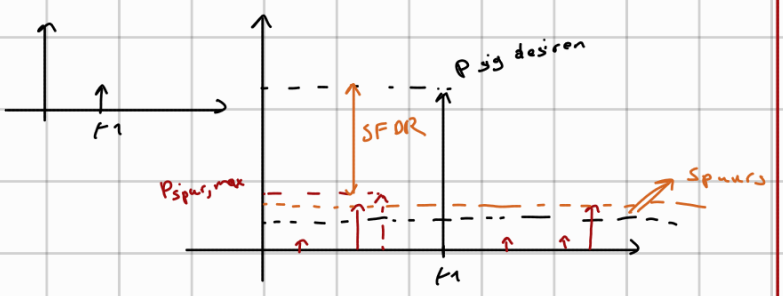


$$DR = IP_{1dB} - P_{sig,in}(min)\Big|_{dB} = IP_{1dB} - MDS$$

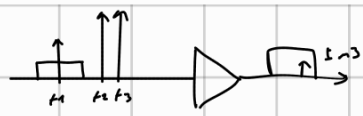
$P_{sig,in}(min)$: MDS — Minimum detectable signal

SFDR

Spur-free Dynamic Range



$$SFDR = P_{sig,desired} - P_{spur(max)}$$



$$SFDR = \frac{2(P_{IIP3} + 174 - NF - 10\log B)}{3} - SNR_{out(min)}$$

